

Confidence interval for difference of two means, dependent samples

Weight loss example, lbs

Background The 365 team has developed a diet and an exercise program for losing weight. It seems that it works like a charm. However, you are interested in how much weight are you likely to lose. You have a sample of 10 people who have already completed the 12-week program. The second sheet in shows the data in kg, if you feel more comfortable using kg as a unit of measurement

Task 1 Calculate the mean and standard deviation of the dataset

Task 2 Determine the appropriate statistic to use

Task 3 Calculate the 95% confidence interval

Task 4 Interpret the result

Optional You can try to calculate the 90% and 99% confidence intervals to see the difference. There is no solution provided for these cases.

Solution:

Subject	Weight before (lbs)	Weight after (lbs)	Difference
1	228.58	204.74	-23.83
2	244.01	223.95	-20.06
3	262.46	232.94	-29.52
4	224.32	212.04	-12.28
5	202.14	191.74	-10.41
6	246.98	233.47	-13.51
7	195.86	177.60	-18.25
8	231.88	213.85	-18.03
9	243.32	218.85	-24.47
10	266.74	236.86	-29.87

Task 1: **Mean** -20.02
 St. deviation 6.86

Task 2: Population variance is unknown
 We have a small sample
 We assume that the population is normally distributed
 The appropriate statistic to use is the t-statistic

Task 3:

95% CI, $t_{9,0.025}$ 2.26

	T	CI low	CI high
95%	-24.93		-15.12

Task 4: You are 95% confident that you will lose between 24.93lbs and 15.12lbs, given that you follow the program as strict as the sample

Note that the solution is exactly the same no matter the unit of measurement